

AI FUNDAMENTALS

AI → Machines performing tasks that normally req. human intelligence.

Speech reco., Lang. understanding,
Decision making.

Types of AI

<u>Narrow AI</u>	<u>General AI</u>
Designed for specific tasks → Image Reco, Translation, Games	Human like intelligence. Can perform many tasks across domains.

Key AI Concepts

1) Machine Learning (ML)

- Algo. learn from data.
- Improves performance over time.
- Used for prediction & pattern detection.

2) Neural Networks

- Inspired by human brain
- Uses connected nodes (neurons)
- Used in facial recog, forecasting

3) Deep learning

- Subset of ML using big neural network
- Handles large datasets
- High comp. appls req.

AI in Cybersec

1) Threat detection

Finds anomalies in logs & traffic.

2) Incident Response

Identifies malware behaviour.

Helps stop attackers early

3) Vulnerability Management

- Supplies risk prioritizations
- Identifies CVEs & attack paths

AI Data Ethics & Responsible Use

AI uses large amounts of data.

Concern: Protect Data Confidentiality
Integrity & Availability [CIAID]

Key Risks to AI

1) Data Breaches

Sensitive data used by AI can be stolen.

2) Biased training data

AI may give unfair or discriminatory results.

3) Malicious attacks

Attackers can manipulate AI inputs / data to cause wrong decisions.

4) Unauthorized access

Poorly controlled AI may reveal protected info.

5) Misuse of Data

Personal data may be used unlawfully or for fraud / identity theft.

Regulations of Laws

(Ex^y) 1)

GDPR - General Data Protection Reg.

- i) Protects personal data
- ii) Covers automated decision making

2) CCPA / CPRA - California Consumer Privacy Act

- i) Protects consumer privacy
- ii) Includes sensitive data like biometric & geolocation data.

3) AI specific laws / proposals
Aim to improve transparency & fairness.

Ethical Concerns

Privacy

Businesses must protect customer data.

AI should not expose or misuse personal info.

Misinformation

Biased or tampered data can produce false info.

Can, trust, create damage

Social division or influence opinion.

Responsible AI Use

1) Use fair and unbiased training data.

2) Protect AI data with strong security.

3) Follow privacy & ethical guidelines.

4) Monitor AI outputs.

AI Emerging Threats

new risks caused by misuse of AI.

Main AI Threats

1) Deepfakes.

4) Privacy violations.

2) Bias & Discrimination.

5) AI - enabled cyberattacks.

3) Autonomous weapons

AI systems that can select / engage targets without human.

Raises ethical legal concerns.

AI can automate phishing, ransomware and attacks.

Attacks Against AI

- | | |
|---|--|
| 1) Prompt Injection
sneak pasted info. | 2) Backdoors in
training data. |
| 3) Data poisoning.
attackers manipulate
training data. | 4) Malicious code or
hidden behavior inserted
in AI training data. |
| 5) Model inversion /
Data inference
attackers try to
extract sensitive data
from AI models. | |

Real world Example: OpenCV

Attacker $\rightarrow$ malicious images $\rightarrow$ exploited OS.
patch in Sept. 2022.

Mitigations

- i) Train employees on AI risk.
- ii) Strong controls and audits.
- iii) Follow ethical and legal guidelines.

TENSORFLOW for ML

Types of ML

<u>Supervised Learning</u>	<u>Unsupervised Learning</u>
Uses labelled data.	Uses unlabelled data.
Input has correct output.	Finds patterns / groups itself.

TensorFlow

free, open source ML platform.
Created for handling data in the
form of tensors.

Tensor

Multi-dimensional array of data.

Eg:-

Vector = 1D

Matrix = 2D

Tensor = 3D or more

Used for images, videos.

Jupyter notebook

write, run, test code in browser.

Uses cells:-

[] = not run, [*] = running
[#] = commented / run order.

Image Classification

AI identifies objects/patterns in images
→ Part of computer vision.

Key Steps in Image Class.

- 1) Data Acquisition
Collect & prepare image dataset.
- 2) Feature Extraction
Identify features like edges, shape, textures.
- 3) Classification
Assign image to a category/label.

CNN Convolutional Neural Network

→ Deep learning model used for Image Recog.
→ Extracts img. features.

CNN Layers

Convolutional Layer → Activation functions

→ Pooling layer → Fully connected layer

→ output layer

How Attackers Exploit Img. Class.

- 1) Exploit Vulnerabilities
eg: → buffer overflow to run malicious code.
- 2) Adversarial Images
Small image changes trick AI into wrong classification.
- 3) Privacy Infringement.
Img. recognition can track people/location.

Generative AI models

1) GANs

Two parts:—

- i) Generator creates fake/noise data.
- ii) Discriminator checks if it is real or fake.
- iii) Used for realistic image/content gen.

2) VAEs

- i) Encode data into key features.
- ii) Decode features to create new, repaired outputs.

3) Transformers

- i) learn patterns in sequence.
- ii) Commonly used for lang. tasks.

4) APTs

- Pre-trained on large text data.
- Type of transformed.
- used for chatbots, writing, coding.

Prompt Injection Attacks

- i) Tricking an AI model into ignoring rules or security restrictions by input.

ii) Attackers manipulate.

Risks / Impact

- 1) Data Leakage
- 2) Unauthorized access
- 3) Misinformation
- 4) Service Disruption
- 5) Reputation Damage.

Why it happens

- i) AI follows patterns and instructions from input.
- ii) If safeguards are weak, malicious prompts can override intended behaviour.

Mitigations

- 1) Use input validation
- 2) Apply context-aware filtering.
- 3) Avoid sending sensitive data to AI.
- 4) Monitor and log AI interaction.
- 5) Regularly update and fine tune models.

AI for Incident Response

- helps analysts detect, investigate and respond to incidents faster.
- AI can automate repetitive tasks.

Suspicious Traffic Analysis

AI/ML can analyze network traffic to detect malicious activity.

Key steps:-

- i) Collect malicious traffic datasets.
- ii) Clean & preprocess the data.
- iii) Extract features like: IP, ports, protocols, packet size.
- iv) Label data as normal or malicious.
- v) Train data & eval. ML models.
- vi) Deploy
- vii) Feedback.